

WASTE MANAGEMENT

Name of the Team Members:

Regd. No:	Name of the student
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Objective:

The objective of a Waste Management System is to efficiently collect, transport, process, recycle, and dispose of waste while minimizing environmental impact and promoting sustainability.

List of Components:

(Also list the components that are used - also mention items that are taken from the kit and used from the lab (such as acrylic, plywood, nails etc) without mentioning the cost)

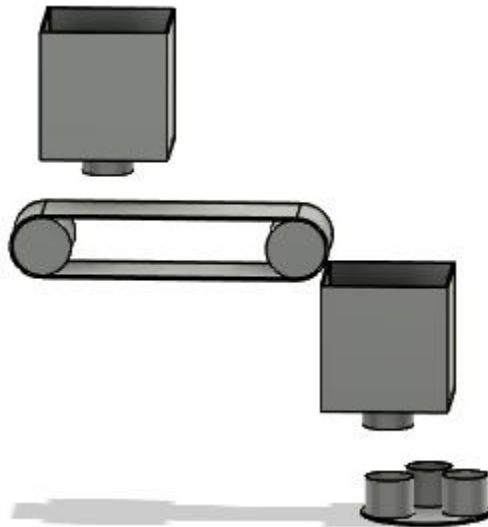
Name of Component (including specifications)	Cost of 1 unit	No. of units	Total cost
IR SENSOR	KIT	1	
MOISTURE SENSOR	KIT	1	
INDUCTIVE PROXIMITY SENSOR	400	1	400
STEPPER MOTOR	150	2	300
WHEEL	50	1	
ARDUINO	KIT	2	
BREAD BOARD	KIT	1	
SERVO MOTOR	KIT	2	

List of Manufacturing Operations & Processes used:

Name of Component	Material	Manufacturing Operation used
Wooded base	wood	Made by tools
Convery belt	Plywood, leather	Made by tools
Wooden box	Wood	Made by tools
3D disc	Plastic, cardboard	Made by tools

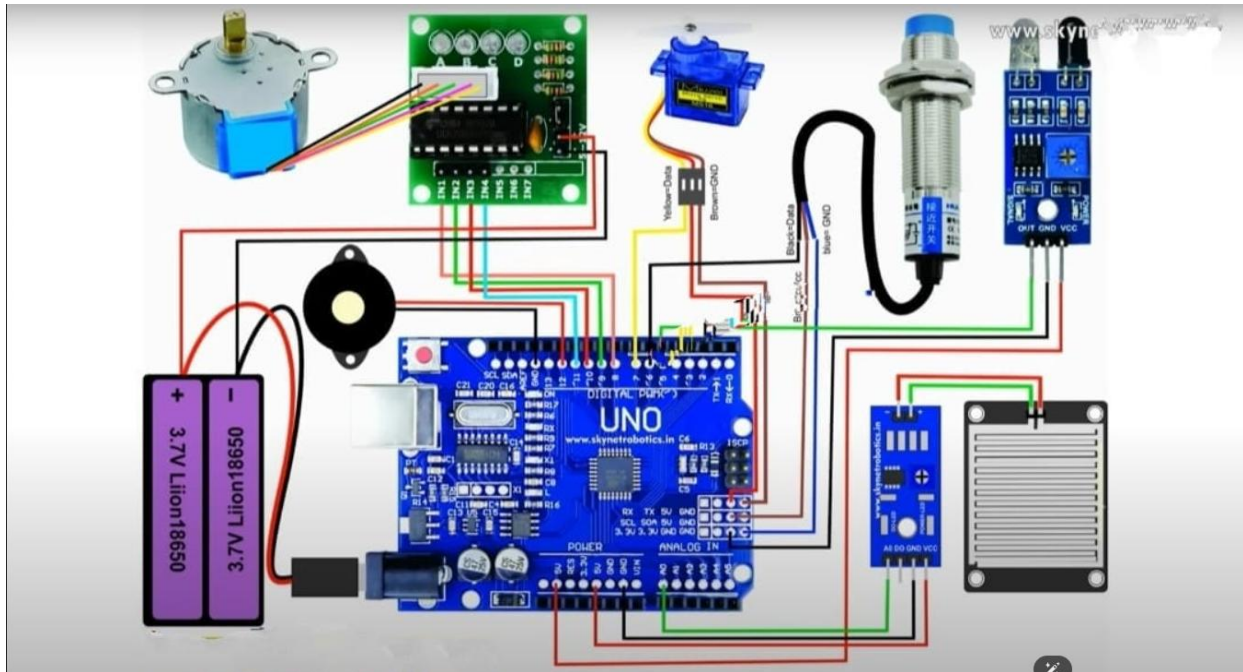
3D Models in Fusion

<https://onedrive.live.com/?qt=allmyphotos&photosData=%2Fshare%2F84AB5442D4E0E72C%21seda11fe76893407c9eabdd3cc04d0819%3Fithint%3Dvideo%26migratedtospo%3Dtrue&sw=bypassConfig&cid=84AB5442D4E0E72C&id=84AB5442D4E0E72C%21seda11fe76893407c9eabdd3cc04d0819&redeem=aHR0cHM6Ly8xZHJ2Lm1zL3YvYy84NGFiNTQ0MmQ0ZTBINzJjL0VlY2ZvZTJUYUh4QW5xdmRQTUJOQ0JrQjY1MEN6QUN2cVIVaEZ0ZTY1Q2IMRFE&v=photos>



Electronic Circuit in Tinkercad

(Give the circuit (or circuits) designed as part of the project):



Description of the Project:

The waste is first dropped into a bin, with the help of servo motor it falls onto a conveyor belt. From conveyor belt (using stepper motor) based on the time delay it falls into the another box .In that box sensors like (ir sensor, soil moisture, proximity) are placed which detects the type of waste and with the help of servo motor opens the lid. Based on this detection, the disc rotates (using the steppermotor) accordingly, ensuring that the waste is directed to its respective bin.

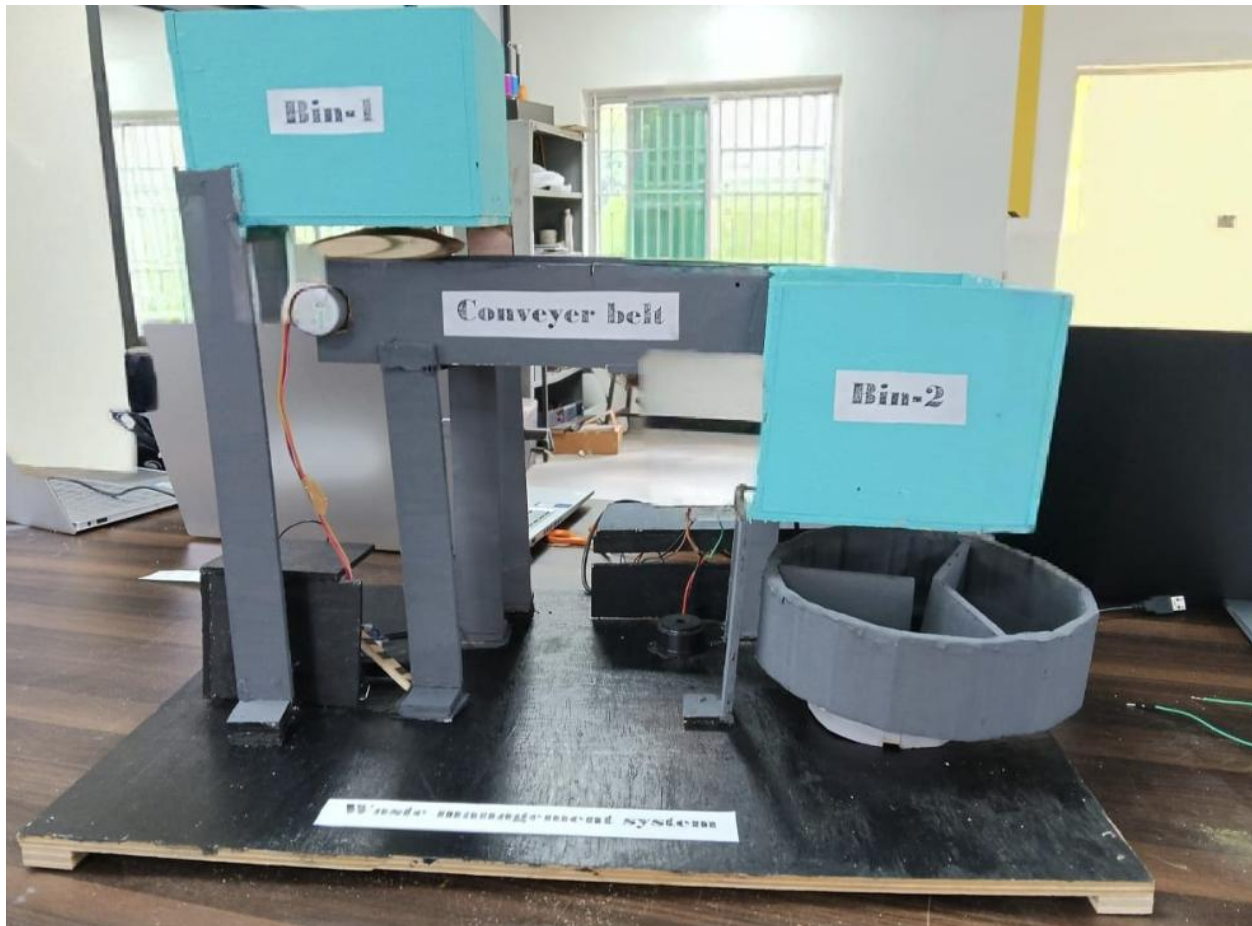
Working Stills:

(Images of the team working)



Final Product Images:

(Photographs of the final product in different views)



Working Video:

<https://photos.app.goo.gl/yxGgWitg1oqKnEmGA>

APPENDIX

(Paste the Arduino codes used in the project)

CODE FOR BIN 1:

```
#include <Servo.h>
```

```
// Define control pins for ULN2003AN driver
```

```
#define IN1 8
```

```
#define IN2 9
```

```
#define IN3 10
```

```
#define IN4 11
```

```
// Define servo pin
```

```
#define SERVO1_PIN 5
```

```
// Steps sequence for 28BYJ-48 (Half-step mode)
```

```
const int stepSequence[8][4] = {  
  {1, 0, 0, 0},  
  {1, 1, 0, 0},  
  {0, 1, 0, 0},  
  {0, 1, 1, 0},  
  {0, 0, 1, 0},  
  {0, 0, 1, 1},  
  {0, 0, 0, 1},  
  {1, 0, 0, 1}  
};
```

```
int stepIndex = 0;
```

```
int direction = -1; // -1 for anticlockwise, 1 for clockwise
```

```
unsigned long lastStepTime = 0;
```

```
unsigned long stepInterval = 2; // milliseconds between steps
```

```
Servo servo1;
```

```
bool pushing = false;
```

```
unsigned long pushStartTime = 0;
```

```
unsigned long pushDuration = 500; // ms
```

```
int stepsCounter = 0;
```

```
int stepsPerRevolution = 512; // Adjust if needed
```

```
unsigned long lastPushCheck = 0;
```

```
void setup() {
```

```
  pinMode(IN1, OUTPUT);
```

```
  pinMode(IN2, OUTPUT);
```

```
  pinMode(IN3, OUTPUT);
```

```
  pinMode(IN4, OUTPUT);
```

```
  servo1.attach(SERVO1_PIN);
```

```
  servo1.write(30); // Neutral
```

```
}
```

```
void loop() {
```

```
  unsigned long currentTime = millis();
```

```

// Stepper motor non-blocking step
if (currentTime - lastStepTime >= stepInterval) {
    lastStepTime = currentTime;
    stepMotor(direction);
    stepsCounter++;

    // Check if it's time to push object
    if (stepsCounter >= stepsPerRevolution && !pushing) {
        stepsCounter = 0;
        servo1.write(110); // Push
        pushing = true;
        pushStartTime = currentTime;
    }
}

// Handle servo return to neutral
if (pushing && (currentTime - pushStartTime >= pushDuration)) {
    servo1.write(30); // Return to neutral
    delay(1000);
    pushing = false;
}

void stepMotor(int dir) {
    stepIndex += dir;
    if (stepIndex > 7) stepIndex = 0;
    if (stepIndex < 0) stepIndex = 7;

    digitalWrite(IN1, stepSequence[stepIndex][0]);
    digitalWrite(IN2, stepSequence[stepIndex][1]);
    digitalWrite(IN3, stepSequence[stepIndex][2]);
    digitalWrite(IN4, stepSequence[stepIndex][3]);
}

```

CODE FOR PIN 2:

```

#include <CheapStepper.h>
#include <Servo.h>

// Define pins
#define IR_SENSOR 2
#define PROXIMITY_SENSOR 3

```

```

#define BUZZER 12
#define SOIL_SENSOR A0
#define SERVO2_PIN 6 // Bin lid servo

// Initialize components
Servo servo2; // Bin opener
CheapStepper stepper(8, 9, 10, 11); // Stepper IN1-IN4

// Track current stepper angle
int currentAngle = 0;

void setup() {
  Serial.begin(9600);

  // Attach bin lid servo
  servo2.attach(SERVO2_PIN);

  // Setup pins
  pinMode(IR_SENSOR, INPUT);
  pinMode(PROXIMITY_SENSOR, INPUT);
  pinMode(BUZZER, OUTPUT);

  // Set stepper speed
  stepper.setRpm(17);
}

void loop() {
  // Check for object with IR sensor
  if (digitalRead(IR_SENSOR) == LOW) {
    Serial.println("Object Detected");
    tone(BUZZER, 1000, 300);
    delay(500);
    int proximityState = 1;
    // proximityState = digitalRead(PROXIMITY_SENSOR);
    Serial.print("Proximity: ");
    Serial.println(proximityState);

    if (proximityState == LOW) {
      // Metal detected
      Serial.println("Metal Detected");
      rotateToBin(0); // Metal bin at 0°
    }
  }
}

```



```

    openBin();
} else {
    // Check soil moisture
    Serial.print("In Soil Moisture: ");
    int moisture = readSoilMoisture();
    Serial.print("Soil Moisture: ");
    Serial.print(moisture);
    Serial.println("%");

    if (moisture > 300) {
        // Wet object
        Serial.println("Wet Waste");
        rotateToBin(200); // Wet bin at 200°
        openBin();
    } else {
        // Dry object
        Serial.println("Dry Waste");
        tone(BUZZER, 1500, 500);
        rotateToBin(120); // Dry bin at 120°
        openBin();
    }
}
}

// Opens the bin lid using servo2
void openBin() {
    servo2.write(30);
    delay(1000);
    servo2.write(110);
    delay(1000);
}

// Rotates stepper to the desired angle
void rotateToBin(int targetAngle) {
    int angleToMove = targetAngle - currentAngle;
    if (angleToMove == 0) {
        return; // Already at the correct position
    }

    if (angleToMove > 0) {

```

```
    stepper.moveDegreesCW(angleToMove);
  } else {
    stepper.moveDegreesCCW(-angleToMove);
  }

  currentAngle = targetAngle;
  delay(1000);
}

// Reads average soil moisture (mapped to %)
int readSoilMoisture() {
  int total = 0;
  for (int i = 0; i < 3; i++) {
    int raw = analogRead(SOIL_SENSOR);
    raw = constrain(raw, 485, 1023); // Sensor dry to wet range
    total += map(raw, 485, 1023, 100, 0); // Higher % = wetter
    delay(100);
  }
  return total / 3;
}
```